

Title

From Acquisition to Assurance: A Scalable Software Framework for Automated Quality Control in Multimodal Brain-Body Imaging

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Category

Talk, Non-Experimental Contributions

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Abstract (492/503)**Introduction**

Multimodal brain-body data collection enables rich, integrative investigations of human cognition, behavior, and physiology, but it also introduces substantial methodological and logistical complexity. The simultaneous acquisition of heterogeneous data streams – often involving distinct hardware systems, software interfaces, and human operators – creates numerous opportunities for failures in experimental design, protocol execution and data integrity. In practice, many research teams initiate large-scale data collection without sufficient, continuous safeguards to detect degradations in data quality as they occur. Pilot datasets may initially appear acceptable, encouraging full deployment, only for subtle equipment malfunctions, sensor drift, or inter-experimenter variability to introduce systematic artifacts over time. Critically, when data are collected at scale and under time pressure, such issues frequently go unnoticed until post hoc analysis, at which point remediation is costly or impossible. These failures can result in substantial losses of time, financial resources, and scientific validity.

To address this challenge, we present [MOBI-QC](#), an automated software framework designed to provide continuous, modality-specific quality assessment for multimodal brain and body imaging data. Developed by cognitive neuroscientists with direct experience in large-scale data collection, MOBI-QC ingests multimodal recordings acquired via the Lab Streaming Layer (LSL) and executes a fully automated pipeline for data processing and quality reporting. With minimal user interaction, the software transforms raw multimodal recordings into a comprehensive publication-ready PDF report summarizing data quality metrics across all or a select few collected modalities.

Methods

MOBI-QC currently supports electroencephalography (EEG), eye tracking, peripheral physiology (electrocardiography, respiration, and electrodermal activity), audio, and video data streams. Each modality is processed using a dedicated pipeline tailored to its signal properties and known failure modes. For EEG, the pipeline includes automated detection of ocular and myogenic artifacts, robust channel assessment using the RANSAC algorithm implemented in the pyPrep framework, and independent components analysis (ICA) for data-driven noise removal. Eye tracking quality

assessment includes metrics such as percentage of valid samples, coordinate system consistency, and three-dimensional gaze validity check. Physiological signals are evaluated using measures including signal-to-noise ratio, percentage of valid data, and baseline drift, among others. Comparable modality-specific metrics are computed for audio and video streams with modules that read .wav and .avi files.

Discussion

All quality metrics implemented in MOBI-QC were selected through an extensive review of the peer-reviewed literature, with the goal of maximizing sensitivity to meaningful data degradations while maintaining interpretability for applied research teams. Although users may adjust parameters to adopt more or less conservative thresholds, the entire pipeline can be executed using well validated default settings, enabling standardized quality assessment without requiring expert intervention.

Beyond single-subject reporting, MOBI-QC supports batch processing and generates group-level summaries that characterize the overall health of the dataset. Subject-level metrics are contextualized with the full sample, allowing rapid identification of outliers and emerging quality trends across participants, sessions, or collection sites.

Together, these features position MOBI-QC as a practical and scalable solution for proactive quality assurance in multimodal brain-body imaging, supporting more efficient data collection, improved reproducibility, and higher confidence in scientific claims.

Figure

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Caption:

References